Knowledge – Chapter 1

Propositional logic, inference and resolution

Knowledge-Based Agents

> Agents that reason by operating on internal representations of knowledge.

E.g. People know facts about the world, and using the information that we know, we are able to draw conclusions.

What we would like to do now is take this idea of knowledge and implement it in our AI.

> **E.g. Facts we know:**
> **S-1:** If it didn't rain, Harry visited Hagrid today.
> **S-2:** Harry visited Hagrid or Dumbledore today, but not both.
> **S-3:** Harry visited Dumbledore today.

Based on these statements we can come to conclusions like:

- Harry did not visit Hagrid today.
- It rained today.

Logic

We need to learn a few terms and symbols so that our AI can reason with logic.

Sentence

> An assertion about the world in a knowledge representation language.
> i.e. some way of representing logic inside an AI.

Propositional Logic

> Logic that is based on propositions. A proposition is just a statement about the world.

Proposition Symbols

P Q R – symbols used to represent propositions (logic).

Logical Connectives

Connectives join propositions, so that we can reason more complexly about other facts that might exist.

- **not** → ¬
- **and** → ∧
- **or** → ∨
- **implication** → →
- **biconditional** → ↔

So, what does each of these symbols represent? Each one is defined by a truth table.

Not (¬)

P	¬P
false	true
true	false

And (∧)

P	Q	P ∧ Q
false	false	false
false	true	false
true	false	false
true	true	true

Or (∨)

P	Q	P ∨ Q
false	false	false
false	true	true
true	false	true
true	true	true

Implication (→)

P	Q	P → Q
false	false	true
false	true	true
true	false	false
true	true	true

> When P is false, P → Q is true whatever Q is. We made no claim about what happens when P is not true, so it doesn't matter what Q is if P is false.

> **E.g.**
> If it's raining (true) → I will be indoors (true) ⟹ **true**
> If it's raining (true) → I am not indoors (false) ⟹ **false**

Biconditional (↔)

> **E.g.** It's raining if and only if I will be indoors. If I am indoors, that means it's raining (and if it's raining, I am indoors).

P	Q	P ↔ Q
false	false	true
false	true	false
true	false	false
true	true	true

Now we know what each symbol means. Next we need to understand where we should use them, and how we know whether P and Q are true or false.

Model

> An assignment of a truth value to every propositional symbol (a “possible world”).

> **P:** It is raining
> **Q:** It is Tuesday
> Model: **{P = true, Q = false}** (this is one of the possible models)

If there are n propositions, then there are 2^n possible models.

Knowledge Base, Entailment and Inference

Knowledge Base (KB)

> A set of sentences known by a knowledge-based agent – the sentences our AI knows to be true. The AI uses this knowledge to represent or form conclusions.

Entailment

> **α ⊨ β** (α entails β)
> In every model in which sentence α is true, sentence β is also true. (α and β are just sentences.)

> **E.g.** “It is a Tuesday in January” (α) ⊨ “It is January” (β)

Inference

> The process of deriving new sentences from old ones.

> **E.g.**
> **P:** It is a Tuesday
> **Q:** It is raining
> **R:** Harry will go for a run
>
> **KB** (knowledge base given to the AI): **(P ∧ ¬Q) → R**
> i.e. if it is Tuesday and it is not raining, then Harry will go for a run. (¬Q means “it is not raining”.)
>
> Now our KB also has the information that:
> • P – it is a Tuesday (P is true)
> • ¬Q – it is not raining (Q is false)

Inference. We know P is true and Q is false, so ¬Q is true. Therefore P ∧ ¬Q is true. The KB says (P ∧ ¬Q) → R, and an implication whose left side is true makes the right side true. So **R is true**: Harry will go for a run.

If instead of ¬Q we had Q (it is raining), then P ∧ ¬Q would be false, so the rule would tell us nothing about R and we could not conclude that Harry will go for a run.

Model Checking

What we are trying to do is use the KB we know to figure out whether α (a new piece of knowledge) is true or not. Does **KB ⊨ α**?

To determine if KB ⊨ α:

- Enumerate all possible models (to enumerate = to list things one by one).
- If in every model where KB is true, α is true, then KB entails α (α must be true if the KB is true).
- Otherwise, KB doesn't entail α.

E.g. P: It is a Tuesday · Q: It is raining · R: Harry will go for a run
KB: (P ∧ ¬Q) → R, P, ¬Q (KB is true only in a model where all of these are true)
Query: **R**

P	Q	R	KB
false	false	false	false
false	false	true	false
false	true	false	false
false	true	true	false
true	false	false	false
true	**false**	**true**	**true**
true	true	false	false
true	true	true	false

KB is true in only one model (P true, Q false, R true), and R is true in it. So KB ⊨ R.

Knowledge Engineering

The process of collecting, organising and representing knowledge so that a computer system can use it to solve problems (or make decisions).

E.g. *(collecting and organising)*
P: The patient has a fever
Q: The patient has a headache

A knowledge engineer might represent it as: **P → Q** – "If the patient has a fever, then the patient has a headache."

Clue

People	Rooms	Weapons
Col. Mustard	Ballroom	Knife
Prof. Plum	Kitchen	Revolver
Ms. Scarlet	Library	Wrench

Who committed the murder, with **what weapon** and in **which room**?

How the game works. Imagine a murder happened in a mansion. There are three things you need to discover:

- **Who?** – which suspect committed the murder
- **With what?** – which weapon was used
- **Where?** – in which room the murder happened

So what we do is take one person, one room and one weapon, put them in an envelope, and try to find out what information is in that envelope.

Propositional symbols

mustard	ballroom	knife
plum	kitchen	revolver
scarlet	library	wrench

- (mustard ∨ plum ∨ scarlet)
- (ballroom ∨ kitchen ∨ library)
- (knife ∨ revolver ∨ wrench)

Information we might have:

- ¬plum
- (¬mustard ∨ ¬library ∨ ¬revolver)

Logic Puzzles

Gilderoy, Minerva, Pomona and Horace each belong to a different one of the four houses: Gryffindor, Hufflepuff, Ravenclaw and Slytherin.

- Gilderoy belongs to Gryffindor or Ravenclaw.
- Pomona doesn't belong in Slytherin.
- Minerva belongs to Gryffindor.

Propositional symbols (one for every person–house pair):

	Gryffindor	Hufflepuff	Ravenclaw	Slytherin
Gilderoy	GilderoyGryffindor	GilderoyHufflepuff	GilderoyRavenclaw	GilderoySlytherin
Minerva	MinervaGryffindor	MinervaHufflepuff	MinervaRavenclaw	MinervaSlytherin
Pomona	PomonaGryffindor	PomonaHufflepuff	PomonaRavenclaw	PomonaSlytherin
Horace	HoraceGryffindor	HoraceHufflepuff	HoraceRavenclaw	HoraceSlytherin

- (PomonaSlytherin → ¬PomonaSlytherin)
- (MinervaRavenclaw → ¬GilderoyRavenclaw)
- (GilderoyGryffindor ∨ GilderoyRavenclaw)

Mastermind

- You are given 4 colours: Red, Blue, Green, Yellow.
- You are then asked to predict which one is in the right order. (I know the order; you have to keep guessing.)

> **E.g.**
> You guess: **Red, Blue, Green, Yellow** → I say: 2 are in the right position.
> Another guess: **B, R, G, Y** → I say: none of these (0).
> Another guess: **R, B, Y, G** → I say: all are correct.

Inference Rules

Inference rules are rules that we can apply to the knowledge that already exists, to turn it into a new form of knowledge.

The general way we structure an inference rule is with a horizontal line. Anything **above** the line is a **premise** – something we know to be true. Anything **below** the line is the **conclusion** that we can arrive at after we apply propositional logic.

> **E.g.**
>
> If it's raining, then Harry is inside
> It's raining
> ---
> **Harry is inside**

Modus Ponens

α → β (true)
α (true)

β (true)

If α → β is true and α is true, then β is true.

And Elimination

α ∧ β (α and β are both true)

α (or β) must be true

> **E.g.**
>
> Harry is friends with Ron and Hermione
> ---
> **Harry is friends with Hermione**

We can take either part of an "and" on its own.

Double Negation Elimination

¬(¬α)

α

> **E.g.**
>
> It is not true that Harry did not pass the test
> ---
> **Harry passed the test**

Implication Elimination

α → β

¬α ∨ β

> **E.g.**
>
> If it is raining, then Harry is inside
> ---
> **It is not raining or Harry is inside**

Biconditional Elimination

α ↔ β

(α → β) ∧ (β → α)

> **E.g.**
>
> It's raining if and only if Harry is inside
> ---
> **If it's raining, then Harry is inside,**
> **and if Harry is inside, then it's raining**

De Morgan's Laws

$$\frac{\neg(\alpha \wedge \beta)}{\neg\alpha \vee \neg\beta}$$

E.g.

$$\frac{\text{It is not true that both Harry and Ron passed the test}}{\textbf{Harry did not pass the test or Ron did not pass the test}}$$

$$\frac{\neg(\alpha \vee \beta)}{\neg\alpha \wedge \neg\beta}$$

E.g.

$$\frac{\text{It is not true that Harry or Ron passed the test}}{\textbf{Harry did not pass the test and Ron did not pass the test}}$$

Distributive Property

$$\frac{\alpha \wedge (\beta \vee \gamma)}{(\alpha \wedge \beta) \vee (\alpha \wedge \gamma)} \qquad \frac{\alpha \vee (\beta \wedge \gamma)}{(\alpha \vee \beta) \wedge (\alpha \vee \gamma)}$$

Search Problem

- Initial state
- Actions
- Transition model
- Goal test
- Path cost function

Theorem Proving (a form of search problem)

- **Initial state:** the starting knowledge base – a set of all the sentences that I know to be true.
- **Actions:** inference rules – the inference rules that I can apply at any given time.
- **Transition model:** the new knowledge base after inference – after applying inference rules, the new set of knowledge is the old knowledge plus some additional knowledge that I have been able to draw.
- **Goal test:** check the statement we are trying to prove – i.e. check whether the statement we want is in the knowledge base we have.
- **Path cost function:** the number of steps in the proof – by choosing the inferences that give us our new statement in fewer steps.

Resolution

E.g.

$$\frac{\begin{array}{c}\text{(Ron is in the Great Hall)} \vee \text{(Hermione is in the Library)} \\ \text{Ron is not in the Great Hall}\end{array}}{\textbf{Hermione is in the Library}}$$

The first line is in the KB.

Unit Resolution Rule

$$\frac{P \vee Q \qquad \neg P}{\mathbf{Q}} \qquad\qquad \frac{P \vee Q_1 \vee Q_2 \vee Q_3 \vee \ldots \vee Q_n \qquad \neg P}{\mathbf{Q_1 \vee Q_2 \vee Q_3 \vee \ldots \vee Q_n}}$$

This is quite useful because Q can itself be a longer propositional logic statement. Once we know ¬P, the rest of the statement ($Q_1 \vee \ldots \vee Q_n$) must be true.

> **E.g.**
>
> $$\frac{\text{(Ron is in the Great Hall)} \vee \text{(Hermione is in the Library)} \qquad \text{(Ron is not in the Great Hall)} \vee \text{(Harry is sleeping)}}{\textbf{(Hermione is in the Library)} \vee \textbf{(Harry is sleeping)}}$$
>
> This is the general resolution rule: $P \vee Q, \neg P \vee R \vdash Q \vee R$.